

PROJECT 11

Predictive Maintenance for Hydraulic Press and Hydraulic Test-Rig Systems

Background

Hydraulic presses and power units rely on pumps, valves, coolers, and accumulators whose gradual degradation is hard to see from the outside but shows up clearly in pressure, flow, and temperature sensor readings. A widely used hydraulic test-rig dataset simulates exactly this kind of condition monitoring, with multiple components degrading at different, independent rates — a realistic multi-target predictive-maintenance problem.

Student Objectives

- Conduct a literature survey on condition monitoring of hydraulic systems in manufacturing (presses, injection units, power packs).
- Study Digital Twin concepts for hydraulic circuits and how component-level degradation (cooler, valve, pump, accumulator) is tracked.
- Understand multi-sensor, multi-cycle hydraulic condition-monitoring data and multi-target condition labels.
- Perform EDA on pressure, flow, and temperature sensor cycles across degradation levels of each component.
- Engineer per-cycle statistical features (mean, range, rate of change) from each sensor channel.
- Build at least three classical ML models (e.g., Random Forest, SVM, Gradient Boosting) to classify the condition/degree of degradation of each hydraulic component.
- Compare single-target (one component at a time) and multi-output classification approaches.
- Explain the most influential sensor features for each component using feature importance.
- Recommend a component-wise preventive maintenance schedule and estimate business impact (reduced press downtime, avoided secondary damage).

Expected Deliverables

- Literature review on hydraulic system condition monitoring
- Business problem analysis for hydraulic press/power-unit reliability
- Data understanding (sensor channels, cycle structure, component condition labels)
- EDA across degradation levels per component
- Data pre-processing (cycle segmentation, normalization)
- Feature engineering (per-cycle statistical features)
- Model comparison (single-target and multi-output) across at least 3 classical ML algorithms
- Confusion matrix and ROC-AUC analysis per component
- Feature importance analysis
- Component-wise maintenance recommendations
- Presentation